Probability and Statistics for Data Science

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1 Probability Theory

1.1 Motivation and a first (big) problem

"Probabilities" are theoretical quantities that can be motivated by long-term relative frequencies. Let A be an event and suppose that an experiment is repeated under the same conditions. We denote by $N(A, n)$ the number of occurrences of A among the first n repetitions. Our intuition suggests

$$\frac{N(A,n)}{n} \overset{*}{\underset{n\to\infty}{\longrightarrow}} \mathbb{P}[A]$$

in some suitable sense.
This suggests some of the basic properties that a probability should satisfy. In particular,

$$N(\Omega, n) = n, \qquad N(\emptyset, n) = 0,$$

and, for A, B,

$$N(A \cup B, n) = N(A, n) + N(B, n) - N(A \cap B, n).$$

Likewise,

$$N(A^c, n) = n - N(A, n).$$

Hence one expects

$$\mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B] - \mathbb{P}[A \cap B], \qquad \mathbb{P}[A^c] = 1 - \mathbb{P}[A].$$

However, it is not possible to define probability by

$$\mathbb{P}[A] := \cancel{\lim_{n\to\infty} \frac{N(A,n)(\omega)}{n}}$$

for every possible realization ω. For example, for infinitely many independent fair coin tosses, the sequence consisting only of tails is a possible outcome, although its relative frequency of tails is 1 and not 1/2. We will see below that such exceptional outcomes can have probability zero. The relation between probability and long-term relative frequencies will be made precise later using laws of large numbers.

1.2 Probability spaces, random variables and their laws

Definition 1.1 (Probability space)**.**
A probability space is a triplet $(\Omega, \mathcal{A}, \mathbb{P})$.

- Ω is the *sample space* and is the set of all possible outcomes of an experiment.
- $\mathcal{A}$ is a σ*-algebra*, i.e. $\mathcal{A} \subseteq 2^\Omega$ and
  1. $\Omega \in \mathcal{A}$,

2. $A \in \mathcal{A} \implies A^c \in \mathcal{A}$,
3. $A_1, A_2, \cdots \in \mathcal{A} \implies \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

The elements of $\mathcal{A}$ are called *events*. In particular,

$$\emptyset \in \mathcal{A}, \qquad A \cap B \in \mathcal{A}, \qquad \bigcap_{n=1}^{\infty} A_n \in \mathcal{A}.$$

- $\mathbb{P} : \mathcal{A} \to [0,1]$ is a *probability measure* if
  1. $\mathbb{P}[\Omega] = 1$,
  2. for pairwise disjoint $A_1, A_2, \cdots \in \mathcal{A}$, i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$,

$$\mathbb{P}\Big[\bigcup_{n=1}^{\infty} A_n\Big] = \sum_{n=1}^{\infty} \mathbb{P}[A_n].$$

  We say that $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{A})$.

The σ-algebra specifies which subsets of Ω are events in the model. For finite sample spaces one often uses $\mathcal{A} = 2^{\Omega}$.

Definition 1.2 (Laplace experiment)**.** A finite probability space $(\Omega, 2^{\Omega}, \mathbb{P})$ is called a *Laplace experiment* if all elementary outcomes have the same probability, i.e. for every $\omega \in \Omega$,

$$\mathbb{P}[\{\omega\}] = \frac{1}{|\Omega|}.$$

In this case, for every $A \subseteq \Omega$,

$$\mathbb{P}[A] = \frac{|A|}{|\Omega|}.$$

Proposition 1.3 (Basic properties of probability measures)**.** *Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. Then:*

1. $\mathbb{P}[\emptyset] = 0$ *and* $\mathbb{P}[A^c] = 1 - \mathbb{P}[A]$ *for* $A \in \mathcal{A}$.
2. *If* $A \subseteq B$, *then* $\mathbb{P}[A] \leq \mathbb{P}[B]$.
3. *For* $A, B \in \mathcal{A}$,
$$\mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B] - \mathbb{P}[A \cap B].$$
4. *If* $A_1, \ldots, A_n$ *are pairwise disjoint, then*
$$\mathbb{P}\Big[\bigcup_{k=1}^{n} A_k\Big] = \sum_{k=1}^{n} \mathbb{P}[A_k].$$
5. *If* $A_1 \subseteq A_2 \subseteq \cdots$ *and* $A = \bigcup_{n=1}^{\infty} A_n$, *then*
$$\mathbb{P}[A_n] \uparrow \mathbb{P}[A].$$

6. If $A_1 \supseteq A_2 \supseteq \cdots$ *and* $A = \bigcap_{n=1}^{\infty} A_n$*, then*

$$\mathbb{P}[A_n] \downarrow \mathbb{P}[A].$$

The last two statements are called *continuity from below* and *continuity from above*. They are used when an event is obtained as an increasing union or a decreasing intersection of events. In particular, they allow probabilities of limiting events to be computed as limits of probabilities.

Definition 1.4. An event $N \in \mathcal{A}$ is called a *null set* if $\mathbb{P}[N] = 0$.
An event $A \in \mathcal{A}$ is said to occur *almost surely* if $\mathbb{P}[A] = 1$.

In general, $\mathbb{P}[N] = 0$ does not imply $N = \emptyset$, and $\mathbb{P}[A] = 1$ does not imply $A = \Omega$.

Definition 1.5 (Measurable space)**.** Let $\mathcal{R}$ be a set and let $\mathcal{B} \subset 2^{\Omega}$. $(\mathcal{R}, \mathcal{B})$ is called a *measurable space* if $\mathcal{B}$ is a σ-algebra.

If $(\Omega, \mathcal{A}, \mathbb{P})$ is a probability space, then $(\Omega, \mathcal{A})$ is a measurable space.

Definition 1.6 (Random variable)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and $(\mathcal{R}, \mathcal{B})$ a measurable space. A *random variable* is a measurable map

$$X : (\Omega, \mathcal{A}) \to (\mathcal{R}, \mathcal{B}),$$

i.e. $X^{-1}(B) \in \mathcal{A}$ for every $B \in \mathcal{B}$. For $\omega \in \Omega$, $X(\omega)$ is called a *realization* of X.

For $B \in \mathcal{B}$ we use the notation

$$[X \in B] := X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}.$$

For a real-valued random variable we use $(\mathcal{R}, \mathcal{B}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, where $\mathcal{B}(\mathbb{R})$ denotes the Borel σ-algebra. It is defined as the smallest σ-algebra that contains all interval $(a, b]$, $a < b$.
The same definition also applies to random vectors, random functions, random matrices, sequences, and other random objects, provided that the target space is equipped with a suitable σ-algebra.

Definition 1.7 (σ-algebra generated by a random variable)**.** Let $X : (\Omega, \mathcal{A}) \to (\mathcal{R}, \mathcal{B})$ be a random variable. The σ-algebra generated by X is

$$\sigma(X) := \{X^{-1}(B) : B \in \mathcal{B}\}.$$

It is the smallest σ-algebra on Ω with respect to which X is measurable.

The σ-algebra $\sigma(X)$ contains the events that can be decided from the observed value of X. If $Y = g(X)$ for a measurable function g, then

$$\sigma(Y) \subseteq \sigma(X).$$

Lemma 1.8 (Doob–Dynkin lemma)**.** *Let $X : (\Omega, \mathcal{A}) \to (\mathcal{R}, \mathcal{B})$ be a random variable and let $Y : (\Omega, \mathcal{A}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a real-valued random variable. Then*

$$\sigma(Y) \subseteq \sigma(X)$$

if and only if there exists a $\mathcal{B}/\mathcal{B}(\mathbb{R})$-measurable function $g : \mathcal{R} \to \mathbb{R}$ such that

$$Y = g(X).$$

Thus, for real-valued Y, the statement that Y contains no information beyond X is equivalent to saying that Y is a measurable function of X.

Definition 1.9 (Law of a random variable)**.** Let $X : (\Omega, \mathcal{A}, \mathbb{P}) \to (\mathcal{R}, \mathcal{B})$ be a random variable. The *law* or *distribution* of X is the probability measure $\mathbb{P}_X$ on $(\mathcal{R}, \mathcal{B})$ defined by

$$\mathbb{P}_X(B) := \mathbb{P}[X \in B] = \mathbb{P}[X^{-1}(B)], \qquad B \in \mathcal{B}.$$

The random variable X and its law $\mathbb{P}_X$ are different objects. The random variable specifies how an observation is obtained from the underlying outcome ω, whereas the law records the probabilities of possible observed values.

1.3 Distribution functions, mass functions and densities

Definition 1.10 (Cumulative distribution function)**.** Let X be a real-valued random variable. The function

$$F_X(x) := \mathbb{P}[X \leq x] = \mathbb{P}_X((-\infty, x]) \qquad (x \in \mathbb{R})$$

is called the *cumulative distribution function* (CDF) of X.

Proposition 1.11. *The CDF F_X of a random variable X has the following properties:*

- *F_X is non-decreasing,*
- *F_X is right-continuous,*
- *$\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$.*

For $a < b$,

$$\mathbb{P}[a < X \leq b] = F_X(b) - F_X(a).$$

Moreover,

$$\mathbb{P}[X = a] = F_X(a) - F_X(a^-),$$

where $F_X(a^-) := \lim_{x \uparrow a} F_X(x)$. We call a *an atom of* X, whenever $\mathbb{P}[X = a] > 0$.

Theorem 1.12 (Existence of a random variable with prescribed CDF)**.** *Let* $F : \mathbb{R} \to [0,1]$ *be non-decreasing and right-continuous with*

$$\lim_{x\to-\infty} F(x) = 0, \qquad \lim_{x\to\infty} F(x) = 1.$$

Then there exists a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ *and a real-valued random variable* X *such that* $F_X = F$*.*

Definition 1.13. A random variable which takes values in a finite or countable set is called a *discrete random variable*.

Definition 1.14 (Probability mass function)**.** If X is a discrete random variable with values in a countable set $\mathcal{S}$, its *probability mass function* (PMF) is

$$p_X(s) := \mathbb{P}[X = s] = \mathbb{P}_X(\{s\}) \qquad (s \in \mathcal{S}).$$

For $B \subseteq \mathcal{S}$,

$$\mathbb{P}_X(B) = \sum_{s\in B} p_X(s).$$

For real-valued discrete X,

$$p_X(x) = F_X(x) - F_X(x^-).$$

Definition 1.15 (Probability density)**.** Let X be a real-valued random variable. A measurable function $f_X : \mathbb{R} \to \mathbb{R}_0^+$ is called a *probability density function* (PDF) of X if

$$\mathbb{P}_X(B) = \mathbb{P}[X \in B] = \int_B f_X(x)\,\mathrm{d}x, \qquad B \in \mathcal{B}(\mathbb{R}).$$

In this case,

$$F_X(x) = \int_{-\infty}^{x} f_X(z)\,\mathrm{d}z.$$

If X has a density, then $\mathbb{P}[X = x] = 0$ for every $x \in \mathbb{R}$.

For a real-valued random variable, the CDF always exists and uniquely determines the law. If the law is discrete, it can also be represented by a PMF. If it admits a density, it can also be represented by a PDF. These are representations of the same probability measure $\mathbb{P}_X$.

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